

Documentation on NSM Infrastructure



About NSM Infrastructure:

The National Supercomputing Mission aims at achieving the goals of attaining self-reliance in supercomputing, building the culture of using supercomputing for carrying out R&D and problem-solving in various domains of scientific and technological endeavours, and designing solutions for various societal applications, and positioning the supercomputing ecosystem in the country at a globally competitive level.

The mission envisions the creation of a national infrastructure of supercomputing systems and facilities of various sizes and scales, distributed across the country and seamlessly integrated through the National Knowledge Network.

The systems and facilities created as part of the infrastructure under this mission are divided into three phases of the National Supercomputing Mission: Phase I, Phase II, and Phase III

List Of Completed Sites:



Param Shivay



Param Shakti



Param Brahma



Param Yukti



Param Sanganak

NSM-Infra



Param Pravega



Param Seva



Param Smriti



Param Utkarsh



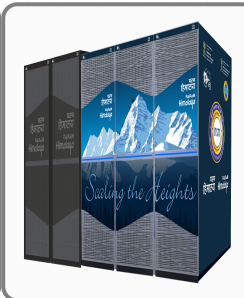
Param Ganga



Param Ananta



Param Porul



Param Himalaya



Param Kamrupa



Param Siddhi-AI



Param Rudra, IUAC



PARAM Rudra, SN Bose



PARAM Rudra, GMRT

How to access NSM HPC System and Policies:

The User Creation portal is designed to streamline the user registration and account creation process. Here's how it works:

1. User Registration:

Users register by providing their official email address, city, and institute name. A verification email is sent to confirm the provided email.

detailed form. Users have the ability to preview and edit the form before submitting it.

3. Document Upload:

Once the registration form is submitted, a link is provided for uploading required documents, such as ID proof, the User Creation Form, and other relevant documents. Please note, after document submission, no further edits can be made, as the documents enter a verification process.

4. Verification & Approval:

The submitted user details and documents are verified by the user's Institute Head of Department (HOD) or Principal Investigator (PI) and Facility/Host Institute. Upon successful verification, the user receives their HPC System credentials.

- Accessing NSM HPC System:

For accessing the NSM hpc system kindly register on [User creation Portal](#) with institutional Email ID

For further information kindly go through the [User Manual](#)

kindly contact nsmsupport@cdac.in for any queries.

- Steps for Allocations:

1. Each NSM site has a Super-computing Allocation committee for Access approval.
2. Users can approach HI/CDAC for NSM resource allocation for HI Internal /NSM Apps/ external users a. Internal user allocation and policies will be decided by Host Institute
3. Users will receive the NSM resource access form
4. External users can approach C-DAC through nsmsupport@cdac.in for resource access.

For Further information kindly visit [NSM resource allocation process](#)

For Infrastructure related information kindly visit [NSM Infrastructure](#)

For HPC Usage Charging Policy:

MSME

Commercial Users

NSM-Infra

Priority	Normal	High	High
CPUcore/hr	Rs. 0.96	Rs. 0.96	Rs. 1.50
GPUcard/hr	Rs. 42	Rs. 42	Rs. 100
Storage/ month	100GB free/account Rs. 1/GB thereafter	100GB free/account Rs. 1/GB thereafter	100GB free/account Rs. 1.50/GB thereafter
Billing Period	Monthly/Quarterly/ Annually	Monthly/Quarterly/ Annually	Monthly/Quarterly/ Annually

Param Pravega Details:



PARAM Pravega, a cutting-edge supercomputing facility, was established under Phase-2 of the National Supercomputing Mission's build approach. It boasts a peak computing power of 3.3 PFLOPS and was designed and commissioned by C-DAC to meet the computational needs of IISc, Bangalore, and various research and engineering institutes in the region. The system is valuable for research in various scientific domains, including Weather & Climate, Oil & Gas, Seismic, Life, Material sciences, and more.

PARAM Pravega - System Configuration

System Specifications

Theoretical Peak Floating-point Performance Total (Rpeak)	3.3 PFLOPS
Base Specifications (Compute Nodes)	2 X Intel Xeon Cascadelake 8268, 24 Cores, 2.9 GHz, Processors per node, 192 GB Memory, 480 GB SSD
Master/Service/Login Nodes	20 nos.
CPU only Compute Nodes (Memory)	300 nos. (192GB)
GPU Compute Nodes (Memory)	40 (192 GB)
GPU ready Compute only Nodes (Memory)	128 nos. (192 GB)
High Memory Compute Nodes	156 nos. (768GB)
Total Memory	245.945 TB
Interconnect	Primary: 100Gbps Mellanox Infiniband Interconnect network 100% non blocking, fat tree topology Secondary: 10G/1G Ethernet Network Management network: 1G Ethernet

CPU Only Compute Nodes

- + 300 Nodes
- + 14400 Cores
- + Compute power of Rpeak 1336.32 TFLOPS
- + Each Node with
 - + 2 X Intel Xeon Cascadelake 8268, 24 cores, 2.9 GHz, processors
 - + 192 GB memory
 - + 480 GB SSD

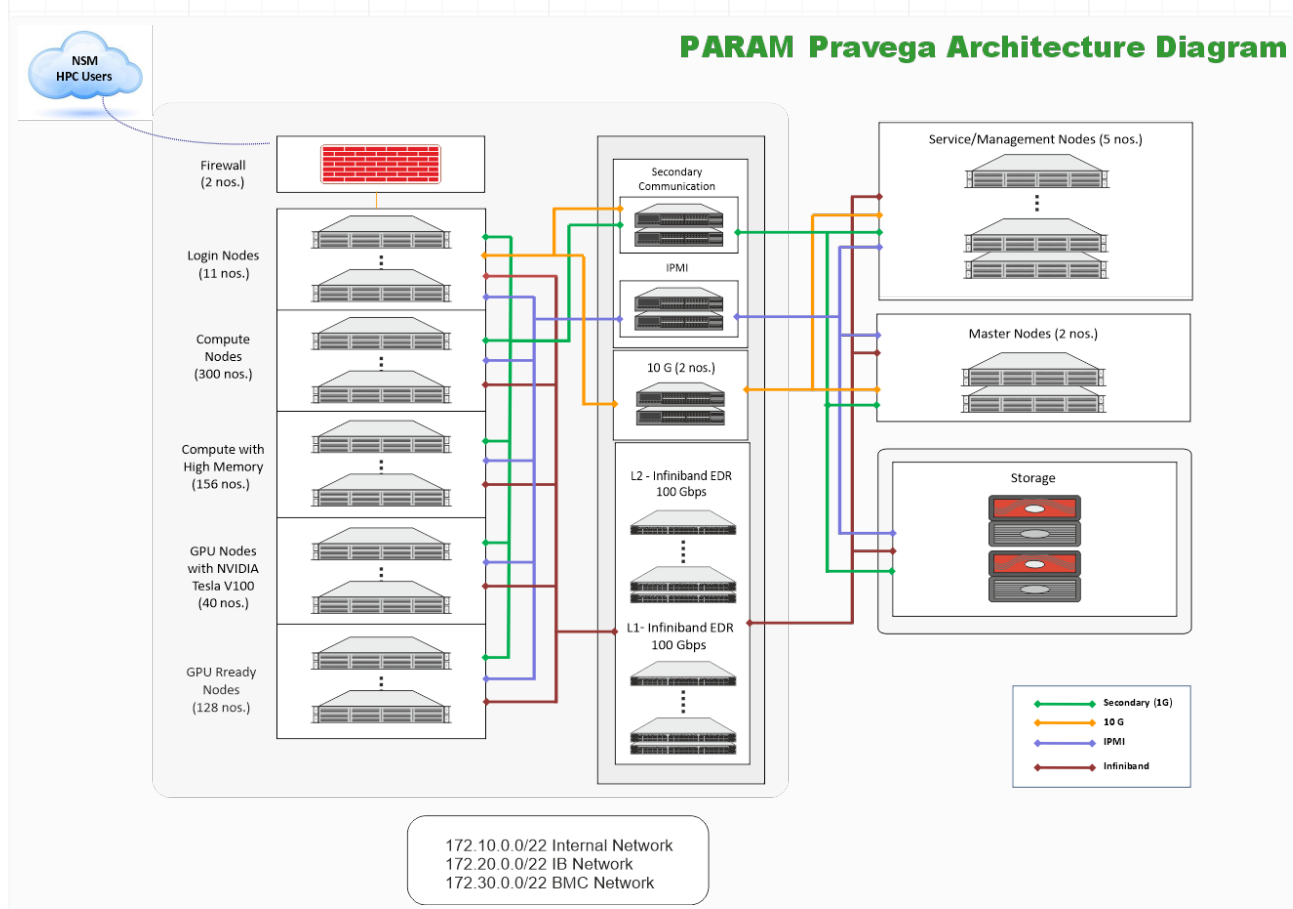
GPU Compute Nodes

- + 40 Nodes
- + 1600 CPU Cores
- + 409600 CUDA Cores
- + Rpeak CPU 128 TFLOPS + GPU 624 TF
- + Each Node with
 - + 2 X Intel Xeon Cascadelake 6248, 20 cores, 2.5 GHz, processors
 - + 192 GB Memory
 - + 2 x NVIDIA V100 SXM2 GPU Cards
 - + 480 GB SSD

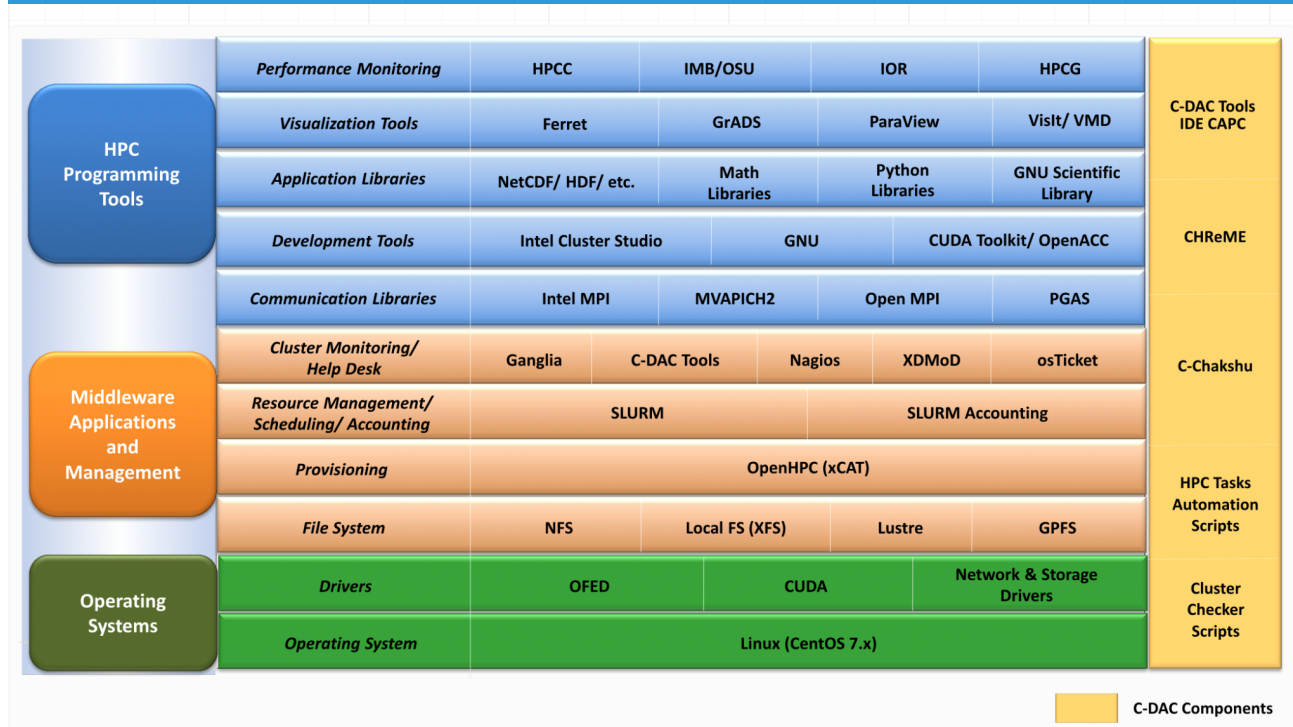
High Memory Compute Nodes

- + 156 Nodes
- + 7488 Cores
- + Compute power of Rpeak 694.88 TFLOPS
- + Each Node with
 - + 2 X Intel Xeon Cascadelake 8268, 24 cores, 2.9 GHz, processors
 - + 768 GB Memory
 - + 480 GB SSD

PARAM Pravega Architecture Diagram



NSM-Infra



Installed Applications/Libraries:

	Bio-informatics	MUMmer, HMMER, MEME, PHYLIP, mpiBLAST, ClustalW
	Molecular Dynamics	NAMD (for CPU and GPU), LAMMPS, GROMACS
HPC Applications	CFD	OpenFOAM, SU2
	Material Modeling, Quantum Chemistry	Quantum-Espresso, Abinit, CP2K, NWChem
	Weather, Ocean, Climate	WRF-ARW, WPS (WRF), ARWPost (WRF), RegCM, MOM, ROMS
Deep Learning Libraries	cuDNN, TensorFlow, Theano	
Dependency Libraries	NetCDF, PNETCDF, Jasper, HDF5, Tcl, Boost, FFTW	

Note: The same details apply to all clusters.

